Market Basket Analysis

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1. Introduction

Market Basket Analysis is used primarily to understand the association between products which is been found out through the frequency of the purchases and also with the joint purchase. The customer basket would reveal all necessary information relating the relationship between different products ie., the antecedent the first product that the customer is choosing then the Consequent the associated product that he demands jointly which is represented with a Confidence ratio to affirm the association.

2. Market Basket Analysis using Groceries_data from Kaggle.

```
  Member_number       Date  itemDescription
1          1808 21-07-2015   tropical fruit
2          2552 05-01-2015       whole milk
3          2300 19-09-2015        pip fruit
4          1187 12-12-2015 other vegetables
5          3037 01-02-2015       whole milk
6          4941 14-02-2015       rolls/buns
```

```
      Member_number       Date       itemDescription
38760          3364 06-05-2014                   oil
38761          4471 08-10-2014         sliced cheese
38762          2022 23-02-2014                 candy
38763          1097 16-04-2014             cake bar
38764          1510 03-12-2014 fruit/vegetable juice
38765          1521 26-12-2014             cat food
```

```
[1] 3898
```

2.1 Data Exploration

The groceries_data set has 38765 observations and 3 features including transaction id(member number), date and Item description. There are 3898 members in this transaction data

```
Apriori

Parameter specification:
 confidence minval smax arem  aval originalSupport maxtime support minlen
        0.2    0.1    1 none FALSE            TRUE       5   5e-04      2
 maxlen target  ext
     10  rules TRUE
```

```
Algorithmic control:
 filter tree heap memopt load sort verbose
    0.1 TRUE TRUE  FALSE TRUE    2    TRUE

Absolute minimum support count: 7

set item appearances ...[0 item(s)] done [0.00s].
set transactions ...[167 item(s), 14963 transaction(s)] done [0.00s].
sorting and recoding items ... [158 item(s)] done [0.00s].
creating transaction tree ... done [0.00s].
checking subsets of size 1 2 3 4 done [0.00s].
writing ... [19 rule(s)] done [0.00s].
creating S4 object  ... done [0.00s].
    lhs                    rhs          support      confidence coverage
[1] {artif. sweetener} => {whole milk} 0.0005346521 0.2758621  0.001938114
[2] {brandy}           => {whole milk} 0.0008688097 0.3421053  0.002539598
    lift     count
[1] 1.746815  8
[2] 2.166281 13
```

We created the rules based on the support (= 0.005), confidence (= 0.2), and minimum length (= 2) using the Apriori algorithm available in the "arules" library. According to the rules created by these specific conditions, if a customer buys "artificial sweetener", there is a 27.58% chance (confidence = 0.2758) for them to buy "Whole milk" also. Similarly, if a customer buys "brandy", there is a 34.21% chance (confidence = 0.3421) for them to buy "Whole milk" as well. However, the support levels for these combinations are extremely low (0.00053 and 0.00086 respectively), indicating that these associations occur very rarely across the entire dataset.

Conclusion

Market Basket Analysis of the Groceries dataset revealed meaningful—though infrequent—associations between certain products. Using the Apriori algorithm with a support threshold of 0.0005 and confidence threshold of 0.2, we identified rules that highlight customer purchasing behavior:

- **Product Associations**: Items like *artificial sweetener* and *brandy* showed moderate confidence levels (27.58% and 34.21%, respectively) in being purchased alongside *whole milk*. This suggests that when these niche items are bought, *whole milk* is a common complementary product.
- **Support vs. Confidence**: While the confidence values indicate a reasonable likelihood of joint purchase, the **low support values** (below 0.001) imply that these combinations are **not widespread** across the entire customer base. These are **niche patterns**, not dominant trends.
- **Strategic Implications**:
    - Retailers could use such insights for **targeted promotions** (e.g., bundling whole milk with artificial sweeteners or brandy).
    - These rules can also inform **store layout decisions**, placing associated items closer together to encourage impulse buys.

About the Author

I'm a student in the MBA Program (2025–2027), currently navigating Trimester I at Amrita School of Business, Amrita Vishwa Vidyapeetham, Coimbatore. This **assignment/blog** was written as part of our coursework for **Introduction to Business Analytics.**

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